Lecture 24

# Clifford algebras and spin representations

**Objectives.** *The half-integral fundamental weights of $B_n$ and $D_n$ do not occur in tensor powers of the vector module $V$. We realise them on an exterior algebra through the Clifford relation.*

By the end of this lecture you will be able to:

- Build the exterior model of the Clifford relation.
- Define the quadratic action of $\mathfrak{so}(V)$ and check its sign.
- Compute weights, parity and dimensions of spin modules.
- Prove irreducibility and identify the highest weights.

Throughout, $V$ is a complex vector space of dimension $N = 2n$ or $N = 2n+1$, with $n \geq 2$. The form $q$ is a nondegenerate symmetric bilinear form on $V$. As in Lecture 23, $\mathfrak{so}(V)$ consists of the maps $A \in \mathfrak{gl}(V)$ with

$$q(Ax, y) = -q(x, Ay) \quad \text{for all } x, y \in V.$$

## 24.1 The Clifford relation and the exterior model

We look for a module whose weights are the $2^n$ vectors $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_n)$, in the coordinates of Lecture 23. An exterior algebra on $n$ vectors suggests one. Its basis is indexed by $n$ independent choices: each index is present or absent. We aim to give these two choices the weights $+\frac{1}{2}\varepsilon_i$ and $-\frac{1}{2}\varepsilon_i$.

This is only a guide, and no proof below uses it. We must still construct operators that satisfy the brackets of $\mathfrak{so}(V)$. First, this section builds operators on an exterior algebra that satisfy the Clifford relation. Next, § 24.2 obtains the action of $\mathfrak{so}(V)$ from quadratic Clifford elements. Finally, §§ 24.3 and 24.4 compute the weights and prove irreducibility.

**Definition 24.1** (Clifford algebra)**.** Let $\mathcal{I} \subseteq T(V)$ be the two-sided ideal generated by all elements $u \otimes v + v \otimes u - 2q(u,v)1$. The *Clifford algebra* is $\mathrm{Cl}(V) = T(V)/\mathcal{I}$. Writing $uv$ for the product of the images of $u, v \in V$,

$$uv + vu = 2q(u,v), \qquad u^2 = q(u,u).$$

A linear map $\gamma$ from $V$ to a unital algebra that satisfies these relations extends to a unital algebra homomorphism on $T(V)$. This homomorphism vanishes on $\mathcal{I}$, so it factors uniquely through $\mathrm{Cl}(V)$.

**Lemma 24.2** (Adapted basis). *The space $V$ has a basis $e_1,\dots,e_n,f_1,\dots,f_n$, together with $u_0$ when $N$ is odd, such that*

$$q(e_i,e_j)=q(f_i,f_j)=0,\qquad q(e_i,f_j)=\frac{1}{2}\delta_{ij},$$
$$q(u_0,e_i)=q(u_0,f_i)=0,\quad q(u_0,u_0)=1.$$

*Proof.* Since $q$ is nondegenerate, diagonalising $q$ and rescaling by complex square roots gives an orthonormal basis $x_1,\dots,x_N$. Put

$$e_j=\frac{1}{2}(x_j+ix_{n+j}),\qquad f_j=\frac{1}{2}(x_j-ix_{n+j}),\qquad u_0=x_{2n+1}\ (\text{odd } N).$$

Then

$$q(e_j,e_j)=q(f_j,f_j)=\frac{1}{4}(1+i^2)=0,\qquad q(e_j,f_j)=\frac{1}{4}(1-i^2)=\frac{1}{2}.$$

Vectors with different indices involve disjoint orthonormal vectors, so $q$ vanishes on them. ∎

Let $W=\operatorname{span}\{e_i\}$ and $W'=\operatorname{span}\{f_i\}$. Both are isotropic of dimension $n$. An isotropic subspace $U$ satisfies $U\subseteq U^{\perp}$, so $2\dim U\le N$. Hence $W$ and $W'$ are maximal isotropic.

The model space is $S=\Lambda W$. For $I=\{i_1<\cdots<i_k\}\subseteq\{1,\dots,n\}$ put

$$e_I=e_{i_1}\wedge\cdots\wedge e_{i_k},\qquad e_\emptyset=1.$$

These $2^n$ vectors form a basis of $S$.

**Definition 24.3** (Creation and annihilation operators). For $w\in W$, the *creation operator* $\mu(w)\colon S\to S$ is given by

$$\mu(w)x=w\wedge x.$$

For $w'\in W'$, the *annihilation operator* $\iota(w')\colon S\to S$ is the linear map with $\iota(w')1=0$ and

$$\iota(w')(w_1\wedge\cdots\wedge w_k)=\sum_{j=1}^{k}(-1)^{j-1}\,2q(w',w_j)\,w_1\wedge\cdots\wedge\widehat{w_j}\wedge\cdots\wedge w_k.$$

Write $\mu_i=\mu(e_i)$ and $\iota_i=\iota(f_i)$.

The right side is multilinear in $w_1,\dots,w_k$. It is also alternating. Suppose $w_a=w_b$ with $a<b$. Terms with $j\neq a,b$ contain a repeated vector. Moving $w_a$ across $b-a-1$ vectors shows that the terms $j=a$ and $j=b$ are one wedge $y$, with coefficients

$$(-1)^{a-1}c\quad\text{and}\quad(-1)^{b-1}(-1)^{b-a-1}c=(-1)^{a}c,\qquad c=2q(w',w_a).$$

They cancel, so $\iota(w')$ is well defined on $\Lambda^kW$. Since $2q(f_i,e_j)=\delta_{ij}$,

$$\iota_i e_I=0\quad(i\notin I),$$
$$\iota_i(e_i\wedge e_J)=e_J\quad(i\notin J).$$

**Proposition 24.4** (Creation and annihilation relations). *For $w,\tilde w\in W$ and $w',w''\in W'$,*

$$\mu(w)\mu(\tilde w)+\mu(\tilde w)\mu(w)=0,$$
$$\iota(w')\iota(w'')+\iota(w'')\iota(w')=0,$$
$$\mu(w)\iota(w')+\iota(w')\mu(w)=2q(w',w)\,\mathrm{id}_S\,.$$

*In particular, $\mu_i\iota_j+\iota_j\mu_i=\delta_{ij}$.*

*Proof.* The first relation holds because $w \wedge \tilde{w} = -\tilde{w} \wedge w$.

For the third, let $x = w_1 \wedge \cdots \wedge w_k$. In $w \wedge x$ the vector $w_j$ sits in position $j+1$. The definition gives

$$\begin{aligned} \iota(w')(w \wedge x) &= 2q(w', w)\, x + \sum_{j=1}^{k} (-1)^j\, 2q(w', w_j)\, w \wedge w_1 \wedge \ldots \widehat{w_j} \cdots \wedge w_k \\ &= 2q(w', w)\, x - w \wedge \iota(w')x. \end{aligned}$$

This also holds for $x = 1$, hence for all $x \in S$ by linearity. It is the third relation.

For the second, let $T = \iota(w')\iota(w'') + \iota(w'')\iota(w')$. Then $T1 = 0$. Put $c' = 2q(w', w)$ and $c'' = 2q(w'', w)$. Two uses of the identity just proved give

$$\begin{aligned} \iota(w')\iota(w'')(w \wedge x) &= \iota(w')\big(c''x - w \wedge \iota(w'')x\big) \\ &= c''\iota(w')x - c'\iota(w'')x + w \wedge \iota(w')\iota(w'')x. \end{aligned}$$

Exchange $w'$ and $w''$ and add. The terms with coefficients $c'$ and $c''$ cancel, so

$$T(w \wedge x) = w \wedge Tx.$$

Induction on degree gives $T = 0$. ∎

**Theorem 24.5** (Exterior model). *Let $\theta \colon S \to S$ be the* parity operator*, given by $\theta x = (-1)^k x$ for $x \in \Lambda^k W$. For $w \in W$, $w' \in W'$ and $t \in \mathbb{C}$ define*

$$\begin{aligned} \gamma(w + w') &= \mu(w) + \iota(w') && (N = 2n), \\ \gamma(w + w' + tu_0) &= \mu(w) + \iota(w') + t\theta && (N = 2n+1). \end{aligned}$$

*Then $\gamma(u)\gamma(v) + \gamma(v)\gamma(u) = 2q(u, v)\,\mathrm{id}_S$ for all $u, v \in V$. Hence $\gamma$ extends to an algebra homomorphism $\gamma \colon \mathrm{Cl}(V) \to \mathrm{End}(S)$.*

*Proof.* Both sides are symmetric and bilinear in $(u, v)$. By polarisation it suffices to prove $\gamma(u)^2 = q(u, u)$. For $u = w + w'$, Proposition 24.4 gives

$$\gamma(u)^2 = \mu(w)^2 + \iota(w')^2 + \mu(w)\iota(w') + \iota(w')\mu(w) = 2q(w', w).$$

Isotropy of $W$ and $W'$ gives

$$q(u, u) = q(w, w) + 2q(w, w') + q(w', w') = 2q(w, w').$$

In the odd case, $\theta$ anticommutes with $\mu(w)$ and $\iota(w')$, because these change degree by one. Also $\theta^2 = 1$. So the cross terms vanish, and

$$\gamma(w + w' + tu_0)^2 = \gamma(w + w')^2 + t^2\theta^2 = 2q(w, w') + t^2 = q(w + w' + tu_0,\, w + w' + tu_0).$$ ∎

## 24.2 Quadratic elements and the action of $\mathfrak{so}(V)$

This section obtains the action of $\mathfrak{so}(V)$ on $S$ from quadratic elements of the Clifford algebra. We state the idea first, and the theorem below proves it. A quadratic Clifford commutator preserves the image of $V$ in $\mathrm{Cl}(V)$ and acts there as a skew map. We name these skew maps $M_{u,v}$, and a map $\sigma$ sends each one back to its quadratic element. Composing $\sigma$ with $\gamma$ gives the Lie algebra action on $S$.

For $u, v \in V$ define $M_{u,v} \in \mathfrak{gl}(V)$ by

$$M_{u,v}x = 2q(v, x)u - 2q(u, x)v.$$

**Lemma 24.6** (Bivectors and skew maps). *Each $M_{u,v}$ lies in $\mathfrak{so}(V)$. The map $u \wedge v \mapsto M_{u,v}$ is a linear isomorphism $\Lambda^2 V \to \mathfrak{so}(V)$. For $A \in \mathfrak{so}(V)$,*

$$[A, M_{u,v}] = M_{Au,v} + M_{u,Av}.$$

*Proof.* The expression

$$q(M_{u,v}x, y) = 2q(v,x)q(u,y) - 2q(u,x)q(v,y)$$

is antisymmetric in $(x, y)$, so $M_{u,v} \in \mathfrak{so}(V)$.

Take an orthonormal basis $x_1, \dots, x_N$. Then

$$M_{x_k,x_l}x_m = 2\delta_{lm}x_k - 2\delta_{km}x_l,$$

so the matrix of $M_{x_k,x_l}$ is $2(E_{kl} - E_{lk})$. In this basis $\mathfrak{so}(V)$ is the space of skew matrices, with basis $E_{kl} - E_{lk}$ for $k < l$. The map sends the basis $x_k \wedge x_l$ $(k < l)$ of $\Lambda^2 V$ to a basis of $\mathfrak{so}(V)$.

For the bracket, use $q(v, Ax) = -q(Av, x)$ and $q(u, Ax) = -q(Au, x)$:

$$\begin{aligned}
[A, M_{u,v}]x &= 2q(v,x)Au - 2q(u,x)Av - 2q(v,Ax)u + 2q(u,Ax)v \\
&= \big(2q(v,x)Au - 2q(Au,x)v\big) + \big(2q(Av,x)u - 2q(u,x)Av\big) \\
&= M_{Au,v}x + M_{u,Av}x. \qquad \blacksquare
\end{aligned}$$

We fix this identification of bivectors with skew maps. Define the linear map $\sigma\colon \mathfrak{so}(V) \to \mathrm{Cl}(V)$ by

$$\sigma(M_{u,v}) = \frac{1}{2}(uv - vu) = uv - q(u,v).$$

It is well defined because the right side is bilinear and alternating in $(u, v)$.

Earlier lectures often write $\rho$ for a representation. Since Lecture 15, however, the bare letter $\rho$ is reserved for the Weyl vector. So in Lectures 24 to 26 we write $\rho_S$ for the action on the spin module $S$. In Lectures 24 and 26 the map $\gamma$ keeps its name. Lecture 25 writes $\rho_S$ for the action of $\mathrm{Cl}(V)$ on $S$, and there $\rho_S \circ c$ is our $\gamma$.

**Theorem 24.7** (The quadratic action). *For $A, B \in \mathfrak{so}(V)$ and $x \in V$, in $\mathrm{Cl}(V)$:*

$$\textit{(a)}\ \ [\sigma(A), x] = Ax, \qquad\qquad \textit{(b)}\ \ [\sigma(A), \sigma(B)] = \sigma([A, B]).$$

*Hence $\rho_S = \gamma \circ \sigma\colon \mathfrak{so}(V) \to \mathfrak{gl}(S)$ is a representation, and $[\rho_S(A), \gamma(x)] = \gamma(Ax)$.*

*Proof.* (a) Apply the Clifford relation twice:

$$\begin{aligned}
uvx &= u\big(-xv + 2q(v,x)\big) \\
&= -\big(-xu + 2q(u,x)\big)v + 2q(v,x)u \\
&= xuv - 2q(u,x)v + 2q(v,x)u.
\end{aligned}$$

Thus

$$[uv, x] = M_{u,v}x.$$

The scalar $q(u, v)$ is central, so $[\sigma(M_{u,v}), x] = M_{u,v}x$. Linearity gives (a) for all $A$.

(b) Let $B = M_{u,v}$. The rule $[a, bc] = [a, b]c + b[a, c]$ and (a) give

$$\begin{aligned}
[\sigma(A), \sigma(B)] &= [\sigma(A), uv] \\
&= [\sigma(A), u]v + u[\sigma(A), v] \\
&= (Au)v + u(Av) \\
&= \sigma(M_{Au,v}) + \sigma(M_{u,Av}) + q(Au, v) + q(u, Av) \\
&= \sigma([A, B]).
\end{aligned}$$

The last step uses the skewness $q(Au, v) = -q(u, Av)$ and Lemma 24.6. Linearity gives (b) for all $B$. The algebra homomorphism $\gamma$ preserves commutators, which gives the final assertions. ∎

We now compute Cartan actions and a bracket. The signs depend on the factor 2 in $M_{u,v}$ and the order $uv$ in $\sigma$. Put $H_i = M_{e_i,f_i}$. Since $2q(f_i, e_j) = \delta_{ij}$,

$$H_i e_j = \delta_{ij} e_j, \qquad H_i f_j = -\delta_{ij} f_j, \qquad H_i u_0 = 0.$$

The definition of $\sigma$ gives

$$\sigma(H_i) = e_i f_i - q(e_i, f_i) = e_i f_i - \frac{1}{2}.$$

On $S$, write $N_i = \mu_i \iota_i$. Then

$$\rho_S(H_i) = \gamma(e_i)\gamma(f_i) - \frac{1}{2} = N_i - \frac{1}{2}.$$

For a bracket, take $A = M_{e_1,e_2}$ and $B = M_{f_1,f_2}$. The definition of $M_{u,v}$ gives

$$A f_1 = -e_2, \qquad A f_2 = e_1.$$

Lemma 24.6 and $M_{f_1,e_1} = -H_1$ give

$$[A, B] = M_{Af_1,f_2} + M_{f_1,Af_2} = M_{-e_2,f_2} + M_{f_1,e_1} = -H_1 - H_2.$$

Since $q(e_1, e_2) = q(f_1, f_2) = 0$,

$$\rho_S(A) = \mu_1 \mu_2, \qquad \rho_S(B) = \iota_1 \iota_2.$$

By Proposition 24.4 with $\delta_{12} = 0$, operators with index 1 anticommute with those with index 2. The same proposition gives $\iota_i \mu_i = 1 - N_i$. Hence

$$\begin{aligned} [\mu_1\mu_2, \iota_1\iota_2] &= \mu_1\mu_2\iota_1\iota_2 - \iota_1\iota_2\mu_1\mu_2 \\ &= -\mu_1\iota_1\mu_2\iota_2 + \iota_1\mu_1\iota_2\mu_2 \\ &= -N_1N_2 + (1-N_1)(1-N_2) = 1 - N_1 - N_2 = \rho_S(-H_1 - H_2). \end{aligned}$$

## 24.3 Weights, parity and dimensions

Let $\mathfrak{h} = \operatorname{span}\{H_1, \dots, H_n\}$, and let $\varepsilon_i \in \mathfrak{h}^*$ satisfy $\varepsilon_i(H_j) = \delta_{ij}$. By the formulas for $H_i$ above, $e_i$, $f_i$ and $u_0$ have weights $\varepsilon_i$, $-\varepsilon_i$ and 0. For weight vectors $x, y$ of weights $a, b$, Lemma 24.6 gives

$$[H, M_{x,y}] = M_{Hx,y} + M_{x,Hy} = (a + b)(H)\, M_{x,y}.$$

The $M_{x,y}$ for unordered pairs of distinct basis vectors form a basis of $\mathfrak{so}(V)$. So $\mathfrak{h}$ is the zero-weight space, and the root vectors are

| vector | $M_{e_i,e_j}$ | $M_{f_i,f_j}$ | $M_{e_i,f_j}$ $(i \neq j)$ | $M_{e_i,u_0}$ | $M_{f_i,u_0}$ |
|---|---|---|---|---|---|
| root | $\varepsilon_i + \varepsilon_j$ | $-\varepsilon_i - \varepsilon_j$ | $\varepsilon_i - \varepsilon_j$ | $\varepsilon_i$ | $-\varepsilon_i$ |
| on $S$ | $\mu_i\mu_j$ | $\iota_i\iota_j$ | $\mu_i\iota_j$ | $\mu_i\theta$ | $\iota_i\theta$ |

with $i < j$ in the first two columns, and the last two columns only for $N$ odd. Each pair has $q(x, y) = 0$, so $\rho_S(M_{x,y}) = \gamma(x)\gamma(y)$. This gives the last row. The positive roots of Lecture 23 are $\varepsilon_i \pm \varepsilon_j$ $(i < j)$, and also $\varepsilon_i$ for $B_n$. Use the fundamental weights of Lecture 23. For $D_2$, set

$$\omega_1 = \frac{1}{2}(\varepsilon_1 - \varepsilon_2), \qquad \omega_2 = \frac{1}{2}(\varepsilon_1 + \varepsilon_2).$$

**Proposition 24.8** (Weights and parity).

*(a) The vector $e_I$ has weight*

$$\nu_I = \frac{1}{2}\Big(\sum_{i\in I}\varepsilon_i - \sum_{i\notin I}\varepsilon_i\Big).$$

*The weights of $S$ are the $2^n$ weights $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_n)$, each of multiplicity one.*

*(b) If $N = 2n$, the spaces*

$$S^+ = \bigoplus_{k\equiv n\ (2)} \Lambda^k W, \qquad S^- = \bigoplus_{k\not\equiv n\ (2)} \Lambda^k W$$

*are submodules of dimension $2^{n-1}$. The weights of $S^+$ have an even number of minus signs. The weights of $S^-$ have an odd number.*

*Proof.* (a) If $i \in I$, write $e_I = \pm e_i \wedge e_{I\setminus\{i\}}$. Then $\iota_i e_I = \pm e_{I\setminus\{i\}}$ and $\mu_i$ restores $e_I$, so $N_i e_I = e_I$. If $i \notin I$, then $\iota_i e_I = 0$. Hence

$$\rho_S(H_i)e_I = \frac{1}{2}e_I \qquad (i \in I),$$
$$\rho_S(H_i)e_I = -\frac{1}{2}e_I \qquad (i \notin I).$$

Distinct subsets give distinct sign patterns.

(b) For basis vectors $x, y \in W \cup W'$, the operator $\rho_S(M_{x,y}) = \gamma(x)\gamma(y) - q(x,y)$ changes degree by $-2$, 0 or 2. These operators span $\rho_S(\mathfrak{so}(V))$, so parity is preserved. The number of minus signs in $\nu_I$ is $n - |I|$, which is even exactly on $S^+$. The map $I \mapsto I\triangle\{1\}$ exchanges even and odd subsets, so each half has $2^{n-1}$ elements. ∎

Figure 24.1 shows these weights for $n = 3$ as the corners of a cube. A root vector with root $\pm\varepsilon_i \pm \varepsilon_j$ changes two signs, so it moves along a face diagonal and stays in one half. For example, $\mu_2\mu_3 e_1 = e_2 \wedge e_3 \wedge e_1 = +e_{123}$. A root vector with root $\pm\varepsilon_i$ exists only in odd dimension. It changes one sign, so it moves along a cube edge and exchanges the two halves.

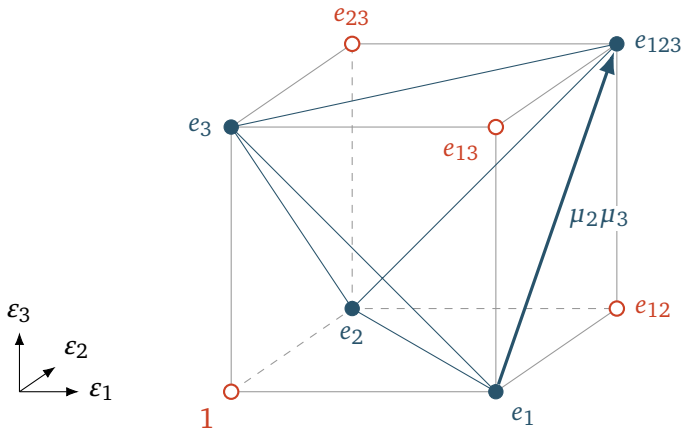



Figure 24.1: The weights $\frac{1}{2}(s_1, s_2, s_3)$ of $S$ for $n = 3$, with $s_i = +1$ exactly when $e_i$ is present. Filled corners form $S^+$ and open corners form $S^-$. The inner lines join the corners of $S^+$ and show the moves by the roots $\pm\varepsilon_i \pm \varepsilon_j$. The arrow is one such move.

The weights of $S$ have coordinates in $\frac{1}{2}+\mathbb{Z}$. Weights of tensor powers of $V$ lie in $\mathbb{Z}^n$ (Lecture 23). Hence no tensor power of $V$ has a submodule isomorphic to $S$ (odd $N$) or $S^{\pm}$ (even $N$).

## 24.4 Irreducibility and highest weights

**Theorem 24.9** (Spin and half-spin modules)**.**

*(a) If $N = 2n$, then $S^+ \cong L(\omega_n)$ and $S^- \cong L(\omega_{n-1})$. Both are irreducible of dimension $2^{n-1}$.*

*(b) If $N = 2n+1$, then $S \cong L(\omega_n)$, irreducible of dimension $2^n$.*

*Proof. Step 1: a weight vector.* Let $U \neq 0$ be a submodule of $S^{\pm}$ (even case) or of $S$ (odd case). The commuting diagonalisable operators $\rho_S(H_i)$ preserve $U$, so $U$ is the sum of its intersections with weight spaces. The weight spaces of $S$ are the lines $\mathbb{C}e_I$. Hence $e_I \in U$ for some $I$.

*Step 2: moving between basis vectors.* The operators in the last row of the root table send each basis vector to $\pm$ a basis vector or to 0:

$$\begin{aligned} \mu_i\mu_j e_I &= \pm e_{I\cup\{i,j\}} && (i\neq j;\ i,j\notin I),\\ \iota_i\iota_j e_I &= \pm e_{I\setminus\{i,j\}} && (i\neq j;\ i,j\in I),\\ \mu_i\iota_j e_I &= \pm e_{(I\setminus\{j\})\cup\{i\}} && (j\in I,\ i\notin I). \end{aligned}$$

In the even case, let $e_J$ be a basis vector with $|J| \equiv |I|$ modulo 2. Removing pairs of indices takes $e_I$ to $\pm e_\emptyset$ or to some $\pm e_{\{k\}}$. In the second case, $|J|$ is odd, so $J \neq \emptyset$. If $k \notin J$, apply $\mu_l\iota_k$ with $l \in J$. Adding pairs then gives $\pm e_J$. Hence $U = S^{\pm}$.

In the odd case, $\iota_i\theta$ removes an index $i \in I$ and $\mu_i\theta$ adds an index $i \notin I$, up to sign. These reach $e_\emptyset$ and then every $e_J$. Hence $U = S$.

*Step 3: highest vectors.* The positive root vectors act by

$$\mu_i\mu_j,\quad \mu_i\iota_j\ \ (i<j), \qquad \mu_i\theta\ \ (B_n \text{ only}).$$

Each operator kills $e_{[n]} = e_1 \wedge \cdots \wedge e_n$. After any contraction, the creation step repeats an index.

On $e_{[n-1]} = e_1 \wedge \cdots \wedge e_{n-1}$, the operator $\mu_i\mu_j$ needs two absent indices, but only $n$ is absent. For $\mu_i\iota_j$, either $j = n$ and $\iota_n e_{[n-1]} = 0$, or $j < n$ and $i$ is still present. So $\mathfrak{n}_+ e_{[n-1]} = 0$ in the even case.

*Step 4: identification.* If an irreducible module $L$ has a highest vector $v$ of weight $\lambda$, the universal property of $M(\lambda)$ gives a nonzero map $M(\lambda) \to L$. It is onto because $L$ is simple. The unique-simple-quotient theorem of Lecture 12 gives $L \cong L(\lambda)$. Apply this to $e_{[n]}$ of weight $\omega_n$, which lies in $S^+$ or in $S$, and to $e_{[n-1]} \in S^-$ of weight $\omega_{n-1}$. Proposition 24.8 gives the dimensions. ∎

## 24.5 Worked example: the half-spin modules of $\mathfrak{so}_6$

Take $n = 3$. Then

$$S^+ = \Lambda^3 W \oplus \Lambda^1 W, \qquad S^- = \Lambda^0 W \oplus \Lambda^2 W.$$

Order the basis of $S^+$ as $(e_{123}, e_1, e_2, e_3)$. Its weights are

$$\begin{aligned} e_{123}&: \ \frac{1}{2}(1,1,1),\\ e_1&: \ \frac{1}{2}(1,-1,-1),\\ e_2&: \ \frac{1}{2}(-1,1,-1),\\ e_3&: \ \frac{1}{2}(-1,-1,1). \end{aligned}$$

The basis $(1, e_{23}, e_{13}, e_{12})$ of $S^-$ has the negatives of these weights, in the same order.

The simple roots are

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \qquad \alpha_2 = \varepsilon_2 - \varepsilon_3, \qquad \alpha_3 = \varepsilon_2 + \varepsilon_3.$$

Since $(\alpha_1, \alpha_2) = (\alpha_1, \alpha_3) = -1$ and $(\alpha_2, \alpha_3) = 0$, the diagram is the $A_3$ chain $\alpha_3 - \alpha_1 - \alpha_2$. On $(\alpha_3, \alpha_1, \alpha_2)$, the labels of $\omega_3$ and $\omega_2$ are $(1,0,0)$ and $(0,0,1)$. These are the two ends of the chain.

The simple root vectors for $\alpha_3, \alpha_1, \alpha_2$ are $M_{e_2,e_3}$, $M_{e_1,f_2}$ and $M_{e_2,f_3}$. On $S^+$ they act by $\mu_2\mu_3$, $\mu_1\iota_2$ and $\mu_2\iota_3$. Their only nonzero values on the basis are

$$\begin{aligned} \mu_2\mu_3 e_1 &= e_2 \wedge e_3 \wedge e_1 = e_{123},\\ \mu_1\iota_2 e_2 &= \mu_1 1 = e_1,\\ \mu_2\iota_3 e_3 &= \mu_2 1 = e_2. \end{aligned}$$

In the ordered basis $(e_{123}, e_1, e_2, e_3)$, the matrices of the raising operators are matrix units of $\mathfrak{sl}_4$. Write $D(A)$ for the matrix of $\rho_S(A)$ in this basis:

$$D(M_{e_2,e_3}) = E_{12}, \qquad D(M_{e_1,f_2}) = E_{23}, \qquad D(M_{e_2,f_3}) = E_{34}.$$

**Proposition 24.10** ($\mathfrak{so}_6$ and $\mathfrak{sl}_4$). *The restriction $\rho_{S^+}\colon \mathfrak{so}(V) \to \mathfrak{gl}(S^+)$ of $\rho_S$ to $S^+$ is a Lie algebra isomorphism. Through it, $S^+$ is the defining $\mathfrak{sl}_4$-module and $S^- \cong (S^+)^*$.*

*Proof.* The algebra $\mathfrak{so}_6$ is simple, since its diagram is connected (Lecture 10). So it equals its derived algebra (Lecture 4), and $\rho_{S^+}$ lands in the traceless operators. Its kernel is a proper ideal, hence zero. Both sides have dimension 15.

The dual $(S^+)^*$ is irreducible. Its weights are the negatives $-\nu$ of the weights $\nu$ of $S^+$. By the weight lists above, these are exactly the weights of $S^-$. The highest weight of $L(\lambda)$ is its unique maximal weight. Equal weight sets therefore give $S^- \cong (S^+)^*$ by Lecture 13. ∎

## 24.6 Conventions in Lectures 23 to 26

Lectures 23 to 26 work with one quadratic space, but they scale their isotropic bases differently. Each lecture is consistent within itself. A formula copied from one lecture into another can still lose a factor. The table below is a reference for such conversions. Lectures 25 and 26 recall their own rows and point back to it.

We compare everything on a common orthonormal basis $x_1, \dots, x_N$ of $V$, as in Lemma 24.2. We write $c(v)$ for the image of $v \in V$ in $\mathrm{Cl}(V)$. For $u, v \in V$, let the skew map $A_{uv}\colon V \to V$ be given by

$$A_{uv}(w) = q(v,w)u - q(u,w)v.$$

| | pairing | isotropic pair | skew map | Clifford map on the pair |
|---|---|---|---|---|
| L23 | $(e_j, e_{-k}) = \delta_{jk}$ | $e_{\pm j} = \frac{1}{\sqrt{2}}(x_j \pm i x_{n+j})$ | $M_{u,v} = A_{uv}$ | none |
| L24 | $q(e_j, f_k) = \frac{1}{2}\delta_{jk}$ | $e_j, f_j = \frac{1}{2}(x_j \pm i x_{n+j})$ | $M_{u,v} = 2A_{uv}$ | $\gamma(e_j) = \mu_j,\ \gamma(f_j) = \iota_j$ |
| L25 | $q(e_j, f_k) = \frac{1}{2}\delta_{jk}$ | $e_j, f_j = \frac{1}{2}(x_j \pm i x_{n+j})$ | $A_{uv}$ | $\rho_S(c(e_j)) = \mu_j,\ \rho_S(c(f_j)) = \iota_j$ |
| L26 | $q(e_j, f_k) = \delta_{jk}$ | $e_j, f_j = \frac{1}{\sqrt{2}}(x_j \pm i x_{n+j})$ | $\varphi(u \wedge v) = A_{uv}$ | $\gamma(e_j) = \sqrt{2}\,\mu_j,\ \gamma(f_j) = \sqrt{2}\,\iota_j$ |

In the row L23, the map $M_{u,v}$ is the one of Exercise 23.6. It equals $A_{uv}$, not $2A_{uv}$, although it has the same letter as here. That lecture writes $(\cdot,\cdot)$ for the form $q$ and $u_j$ for the basis $x_j$.

Superscripts now name the lecture. The skew maps and the map $\sigma$ are related by

$$M^{(24)}_{u,v} = 2A_{uv}, \qquad A^{(25)}_{uv} = \varphi^{(26)}(u \wedge v) = A_{uv}, \qquad \sigma(A_{uv}) = \frac{1}{4}[c(u), c(v)].$$

The isotropic vectors are related by

$$e^{(24)}_j = e^{(25)}_j = \frac{1}{\sqrt{2}} e^{(23)}_j = \frac{1}{\sqrt{2}} e^{(26)}_j,$$

and in the same way for $f_j$, where $f^{(23)}_j$ means $e_{-j}$.

The factor $\sqrt{2}$ in the last row does not give a new operator. Match the wedges $e_I$ of this lecture with those of Lecture 26. Then

$$\gamma^{(26)}\big(e^{(26)}_j\big) = \sqrt{2}\,\mu_j = \gamma^{(24)}\big(\sqrt{2}\,e^{(24)}_j\big),$$

so the two Clifford maps agree on every vector of $V$. The map $\rho_S \circ c$ of Lecture 25 is also this $\gamma$.

A formula must therefore be copied together with its pairing. For example, keep the operators $\mu_j$ and $\iota_j$ of this lecture, which satisfy $\mu_j\iota_j + \iota_j\mu_j = 1$. Then $\gamma(e_j) = \mu_j$ and $\gamma(f_j) = \iota_j$ with the pairing $q(e_j, f_j) = 1$ of Lecture 26 would give $\gamma(e_j)\gamma(f_j) + \gamma(f_j)\gamma(e_j) = 1$. The Clifford relation requires the value 2 there.

A few names also change. In odd dimension the extra unit vector is $x_{2n+1}$. Lectures 23 and 26 call it $e_0$, and Lectures 24 and 25 call it $u_0$. Lecture 25 names the orthonormal basis $u_{2j-1} = x_j$ and $u_{2j} = -x_{n+j}$. Lecture 26 names no orthonormal basis, so its row records how we match its vectors with the $x_j$. It also uses $E$ for the space $W$ of this lecture, and it uses $W$ for the span of the $e_i$ and $f_i$.

## Self-checks

(1) What are the dimensions and highest weights of the spin modules of $\mathfrak{so}_8$ and $\mathfrak{so}_9$?

(2) Why is $e_\emptyset = 1$ not a highest vector of $S$?

(3) Which half-spin module of $\mathfrak{so}_{10}$ contains $e_{13}$, and what is its weight?

*Answers.* (1) $\mathfrak{so}_8$: $S^+ \cong L(\omega_4)$ and $S^- \cong L(\omega_3)$, both of dimension 8. $\mathfrak{so}_9$: $S \cong L(\omega_4)$, of dimension 16. (2) Because $\rho_S(M_{e_1,e_2})1 = \mu_1\mu_2 1 = e_{12} \neq 0$. (3) Here $n = 5$ and $2 \not\equiv 5$ modulo 2, so $e_{13} \in S^-$. Its weight is $\frac{1}{2}(1, -1, 1, -1, -1)$.

## Summary

Creation and annihilation operators on $\Lambda W$ realise the Clifford relation. The elements $\frac{1}{2}(uv - vu)$ give the action of $\mathfrak{so}(V)$, with half-integral weights $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_n)$.

For $D_n$, $S = S^+ \oplus S^- \cong L(\omega_n) \oplus L(\omega_{n-1})$. For $B_n$, $S \cong L(\omega_n)$.

The last section compares the normalisations of Lectures 23 to 26.

### Reading

W. Fulton and J. Harris, *Representation Theory: A First Course*, §20.1. R. Carter, *Lie Algebras of Finite and Affine Type*, §13.5. Goodman–Wallach, *Symmetry, Representations, and Invariants*, Chapter 6.